



## Lab 04: Router Configuration & Subnetting

Switch0 (2960-24TT) · Router0 (2911) · PC0 · PC1 · PC2

Lab #04

Topic: Subnetting & Routing

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

### 1. Lab Objectives

- Connect Switch0 (2960-24TT) to Router0 (2911) via Fa0/3 to Gig0/0.
- Configure the router Gigabit interface as the LAN default gateway.
- Assign static IPs to PC0 and PC1 on the switch subnet.
- Connect PC2 directly to Router0 Gig0/1 as a separate subnet.
- Verify full connectivity using ping and show commands.

### 2. Network Topology

Switch0 connects PC0 (Fa0/1) and PC1 (Fa0/2) and uplinks to Router0 on Fa0/3 → Gig0/0. PC2 plugs directly into Router0 Gig0/1. Two /24 subnets: 192.168.1.0/24 for the switch LAN and 192.168.2.0/24 for PC2.

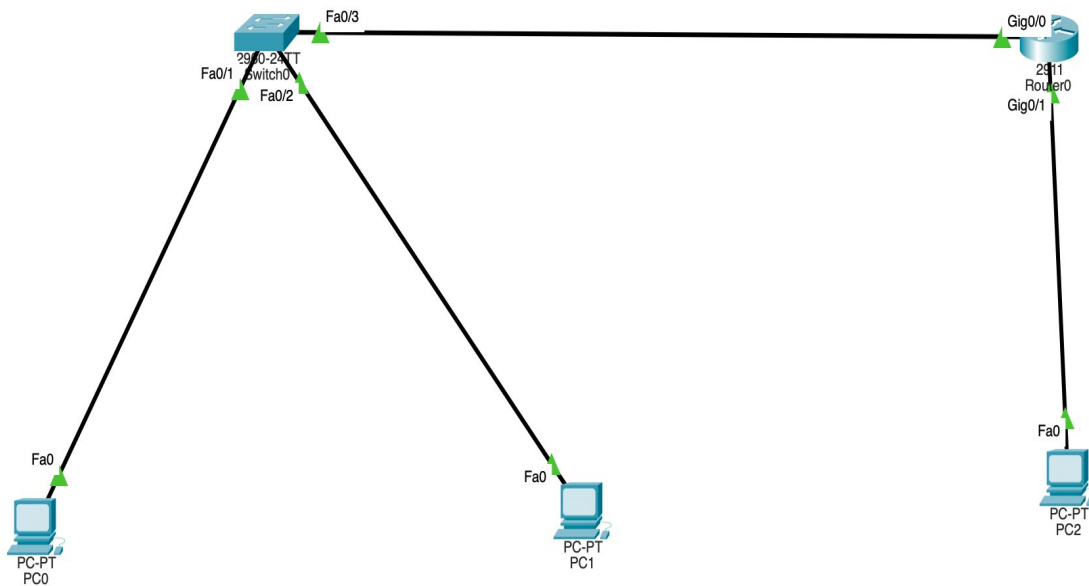


Figure 1 – Switch0 (PC0/PC1) → Router0 (2911) → PC2 direct on Gig0/1

| Device  | Interface | IP Address   | Subnet Mask   | Gateway     |
|---------|-----------|--------------|---------------|-------------|
| Router0 | Gig0/0    | 192.168.1.1  | 255.255.255.0 | —           |
| Router0 | Gig0/1    | 192.168.2.1  | 255.255.255.0 | —           |
| PC0     | Fa0       | 192.168.1.10 | 255.255.255.0 | 192.168.1.1 |



|     |     |              |               |             |
|-----|-----|--------------|---------------|-------------|
| PC1 | Fa0 | 192.168.1.20 | 255.255.255.0 | 192.168.1.1 |
| PC2 | Fa0 | 192.168.2.10 | 255.255.255.0 | 192.168.2.1 |

### 3. Configuration

```
Router>enable
Router#configure terminal
Router(config)#hostname Router0
Router0(config)#interface GigabitEthernet0/0
Router0(config-if)#ip address 192.168.1.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface GigabitEthernet0/1
Router0(config-if)#ip address 192.168.2.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0#write memory
```

### 4. Verification

```
PC0>ping 192.168.1.20
PC0>ping 192.168.2.10
Router0#show ip interface brief
Router0#show ip route
```

### 5. Key Takeaways

- Two router interfaces create two separate subnets automatically.
- PC0 and PC1 on the same subnet communicate via the switch without involving the router.
- PC2 on a different subnet requires the router to forward its traffic.
- Connected routes appear in 'show ip route' automatically when interfaces are up/up.
- The 2960 switch needs no routing config — it forwards frames by MAC address only.